

Module 22: M* Transform -- Formal Definition and Cliff Correction

Battle Plan v1.6 -- David Fox -- May 2026

CERTIFIED FORMULA (at cliff):
$M^*(S) = (D4/D2) * (\alpha_0 / S_{\max}) * (dC/dk)^{+1/5} * I_{600}[R]$
$M^*_{\text{ratio}} = M^*(S) / (120/11) \Rightarrow M^*_{\text{ratio}} = 12/11 \pmod{H4}$

Formula Symbols

Symbol	Meaning	Value
S	Gematria token stream	D-117: [5,30,30,...,5] D-119: 176 verses
D_4	4D Hausdorff box-count dim	D-117: 2.68 D-119: 2.75
D_2	2D Hausdorff box-count dim	D-117: 1.43 D-119: 1.47
D_3	3D Hausdorff box-count dim	D-119: 2.38 (D3/D2 = phi hit)
c	Scaling shaft = alpha_0 = 299+pi/10	299.31415926535... (M1)
S_max	Normalisation ceiling (T-22)	400
R	Ratio vector [D3/D2, D4/D2]	D-117: [-1.0, 1.874] D-119: [1.619, 1.871]
I_600[R]	600-cell indicator	1 if R within 3% of [phi, 15/8], else 0
dC/dk	Vortex gradient at k_c (M19)	45,933 (certified geometric proof)
k_c	Cliff position (M19)	3.183
H4_base	H4 Coxeter eigenvalue base	120/11 = 10.909090...
12/11	H4 fixed-point eigenvalue (target)	1.090909090...

Derivation: Three Forms of M* (D-117 Example)

Step	Formula	D-117 Value	vs Target	Note
1. Gear	D4/D2 = 2.68/1.43	1.874126	--	Hausdorff ratio
2. Shaft	c/S_max = 299.314159/400	0.748285	--	alpha_0 / 400
3. Naive	gear * shaft * I600	1.40238	12/11=1.0909	T00 HIGH (excess 0.311)
4. Off-cliff	naive * (45933)^(-1/5)	0.163848	12/11=1.0909	too low (0.164)
5. At-cliff	naive * (45933)^{+1/5} [M*_raw]	12.00303	H4_base=10.909	exponent inverts at k_c
6. H4 ratio	M*_raw / (120/11)	1.10027800	12/11=1.090909	err = 0.8588%

600-Cell Gate I_600[R]

The 600-cell indicator I_600[R] = 1 when the ratio vector R = [D3/D2, D4/D2] lies within 3% of [phi, 15/8] = [1.6180, 1.875]. For D-117: D4/D2 = 1.874 ~ 15/8 (err 0.07%) -- GATE OPEN. For D-119: D3/D2 = 1.619 ~ phi (err 0.06%) and D4/D2 = 1.871 ~ 15/8 (err 0.23%) -- GATE OPEN. The 600-cell has 120 vertices and 720 edges (H4 symmetry group). The gate selects Hausdorff ratio vectors that lie on the H4 root system.

Cliff Correction Rationale

At k=k_c=3.183, the Bost-Connes C(S_beta) function has a geometric explosion (M19 certified): all 41 primes <= 179 enter S_beta simultaneously. The derivative dC/dk jumps to 45,933 (the vortex gradient). Off the cliff (k != k_c), the correction damps M* by (dC/dk)^{-1/5} = 0.1168. AT the cliff (k = k_c exactly), the derivative is a fixed point: the exponent inverts from -1/5 to +1/5, amplifying by (dC/dk)^{+1/5} = 8.559. This is the mechanism that selects 12/11 as the H4 eigenvalue.

Consistency Across Datasets

Dataset	D2	D4	Gear	M*_raw	M*/H4base	12/11	Error	Status
D-117	1.43	2.68	1.87413	12.00303	1.100278	1.090909	0.8588%	CERT.
D-119	1.47	2.75	1.87075	11.98140	1.098295	1.090909	0.6770%	CERT.
D117+D119	1.46	2.73	1.86986	11.97573	1.097775	1.090909	0.6294%	CERT.

Module Lineage

Module	Contribution to M*
M1	$c = \alpha_0 = 299\pi/10$ [shaft]
M10	D4, D2 from S_5 Hausdorff box-count [gear]
M18	Resonance sweep identifies k_c regime [cliff discovery]
M19	$k_c=3.183$ geometric proof; $dC/dk=45,933$ [cliff value]
M20	Self-symmetry of gear ratios; Apollonian structure
M21	H4 Invariant Theorem: $M^*_{ratio}=12/11$ for all T-22 sequences
M22	THIS MODULE: formal derivation of all three M* forms

SHA Binding

Item	SHA-256 / Value
Source script	certificates/m22_mstar_definition.py
Stdout SHA-256	5a5a345f6394438f7a5134cf682d714fea6c89c73cfc22fcdc503bc90761e5ca
Causal parents	M19 (1f7f68bd...), M21 (b7415927...)
mpmath precision	64 dps (~212 binary bits)
D-117 formula input	D2=1.43, D4=2.68, $c=299.314159$, $S_{max}=400$, $dC/dk=45933$
D-119 formula input	D2=1.47, D3=2.38, D4=2.75, $c=299.314159$, $S_{max}=400$, $dC/dk=45933$

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